Did Generative AI Homogenize Online Writing?

Evidence from 30 Corpora, 2008–2026

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Draft v0.2 — October 20, 2026

Abstract

We test whether online writing became more homogeneous after the public release of ChatGPT (2022Q4). We embed the full text of about 1.2 million sampled answers and abstracts from 30 dated corpora—26 Stack Exchange communities (14 high- and 12 low-cognitive-load) plus Hacker News, PubMed oncology, arXiv computer science and arXiv mathematics—and test, quarter by quarter, for a change in trend at 2022Q4. Four findings. (1) Answers became more similar to each other: the similarity trend turns upward after ChatGPT in 24 of 26 Stack Exchange communities, significantly in 15, and in all 4 external corpora; with a second, independent text encoder the rise holds in expert communities and in all external corpora, but not in everyday ones. (2) The turn is stronger in high-cognitive-load communities, where AI saves more effort (11 of 14 versus 4 of 12, and 9 of 14 versus 2 of 12 with the second encoder; $p \approx 0.002$–0.03). (3) The change is mainly in *what* is asked and discussed: the mix of subjects shifts in 24 of 26 communities and answers within the same subject converge in 18 of 26, but answers to the *same question* converge in only 6. (4) Human answers do not move towards what a language model would write for the same question. We conclude that generative AI is associated with a real but specific form of homogenization—a narrowing and convergence of the questions people bring and the subjects they discuss, strongest in expert communities—rather than people writing like the model.

1 Motivation

Since late 2022 a common complaint has surfaced: *everything online is starting to sound the same.* As hundreds of millions of people route their writing and their questions through a handful of large language models, the worry is that the diversity of human expression—and, more importantly, of *ideas*—collapses towards a shared machine mean.

This matters beyond aesthetics. If the range of answers to a question narrows, so does the pool of ideas a reader meets. Reference sites and scientific abstracts are the durable memory of a field; if that record converges, we lose signal about what people actually discovered or argued. There is also a prior reason to expect an effect: a meta-analysis of human–AI co-creation studies finds a small, persistent homogenization effect [2], and behavioural economics predicts that an always-available, fluent default answer is exactly the kind of anchor people converge on [9].

"Did AI homogenize writing?" is too coarse to test. We split it into four questions, each with its own test:

1. **Did texts become more similar to each other?** Are two answers from the same quarter more similar after ChatGPT than the earlier trend predicts?

2. **Where is it stronger?** If AI saves the most effort on demanding questions, the effect should be larger in high-cognitive-load communities.

3. **What changed?** Texts can become more alike because people discuss a narrower or different *set of subjects*, because answers on the *same subject* converge, or because answers to the *same question* converge. These have different consequences, and we separate them.

4. **Do people write like the model?** Do human answers move towards what a language model would write for the same question?

2 Data

All corpora are public, dated to the quarter, and run to 2026Q2, giving 15 quarters of post-ChatGPT observation against a pre-2022 baseline of 6 to 14 years (Table 1).

Stack Exchange. From the June 2026 data dump we use 26 question-and-answer communities with healthy recent volume: 14 *high-cognitive-load* communities, where questions are technical or expert (Cross Validated, Philosophy, Economics, Cryptography, Law, History, Computer Science, Biology, Astronomy, Chemistry, Electrical Engineering, Physics, Software Engineering, English Language & Usage), and 12 *low-cognitive-load* ones, about everyday, practical or hobby subjects (Seasoned Advice, Travel, Bicycles, Gardening, Role-playing Games, Board Games, Personal Finance, Photography, The Workplace, Home Improvement, Science Fiction & Fantasy, Arqade). The unit is a single answer. Each answer is linked to its question and to the question's tags, which the asker chooses.

External corpora. To check that results are not specific to Stack Exchange's format we add four corpora from other platforms: Hacker News comments, PubMed oncology abstracts, and arXiv computer-science and mathematics abstracts. They have no questions or tags, so they are used for the similarity trend (question 1) and the embedding-topic test (question 3), but not for the cognitive-load comparison, the tag and same-question tests, or the model anchor, which use the 26 Stack Exchange communities only. Table 2 lists the corpora behind each test.

Table 1: Corpora. Up to 800 texts are sampled per quarter (quarters with fewer than 30 are dropped).

Group	Corpora	Texts	Sampled	Span
Stack Exchange, high cognitive load	14	1.80M	0.54M	2008Q3–2026Q2
Stack Exchange, low cognitive load	12	1.08M	0.50M	2009Q4–2026Q2
External (HN, PubMed, arXiv CS, arXiv Math)	4	0.21M	0.13M	2016Q1–2026Q2

3 Method

Turning texts into vectors. A sentence encoder—here all-MiniLM-L6-v2 [8, 11]—maps a passage to a vector of 384 numbers such that passages with similar meaning get vectors pointing in similar directions. The encoder reads at most 128 tokens (about 90 words) at a time, so to represent a whole answer we clean it of URLs, DOIs, e-mail addresses and machine strings, split it into sentences, pack the sentences into chunks of at most 120 tokens, embed each chunk, and average the chunk vectors, weighting each by its length.

Note for Mark (draft only – remove before release)

Draft v0.1 embedded only the first 128 tokens of each text, so it measured how answers *open*; every embedding result in this draft is recomputed on whole texts. Two results changed: the cognitive-load comparison (Section 4.2) now shows a clear difference, where v0.1 found none, and same-subject convergence (Section 4.3) is now found on many sites. This note, and any mention of v0.1, will be removed in the release version.

Measuring how alike texts are. The similarity of two texts is the *cosine similarity* of their vectors: the cosine of the angle between them, from about 0 for unrelated texts to 1 for texts with the same meaning (in practice, two random answers from the same community score about 0.05–0.2). For each corpus and quarter we sample up to 800 texts and take the average cosine similarity over all pairs of them. A rising average means that texts written in the same quarter are becoming more similar to one another.

Testing for a change at ChatGPT. For each corpus we fit a straight trend line to the quarterly similarity before 2022Q4 and another after it (an *interrupted time series*, or segmented regression [10]), with standard errors that allow for correlation between neighbouring quarters [7]. Our main statistic is the *slope change*: the slope of the line after ChatGPT minus the slope before, in similarity units per quarter. A positive, significant slope change means similarity began rising faster (or stopped falling) after ChatGPT; Figure 1 shows an example. The same test is applied to every measure below. Because we test many corpora, we also report how many results survive the Benjamini–Hochberg correction [1], which limits the expected share of false positives among the results called significant to 5%.

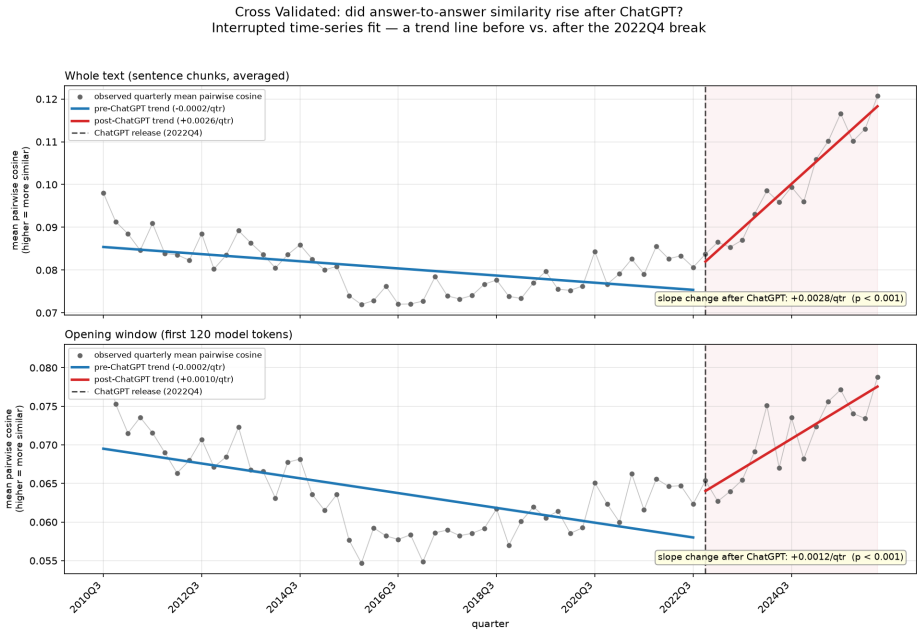


Figure 1: How the slope change is measured, for Cross Validated. Dots: average similarity between answers in each quarter. Blue line: trend before ChatGPT (−0.0002 per quarter); red line: trend after (+0.0026). The slope change is the difference, +0.0028 per quarter. Top: whole answers; bottom: first 120 tokens only (the length check of Section 4.1).

Tests. We call each test a *family*: a measure computed per corpus and quarter, its slope-change test, and its plots, numbered as in the accompanying code. Table 2 maps each question to its families and the corpora they use. Earlier exploratory families are summarised in Appendix A.

Table 2: Questions, tests, and the corpora each test uses.

Question	Family	Test	Corpora
1. More similar?	5, 8	Average similarity between texts each quarter; slope change at 2022Q4	all 30
2. Where stronger?	9	Slope change, scaled by each site's pre-2022 variability, high vs low load	26 SE
3. What changed?	13	Same-subject similarity, number of subjects and subject mix, with subjects = 50 embedding topics	all 30
	14	The same, with subjects = Stack Exchange tags	26 SE
	11	Similarity between answers to the same question	26 SE
4. Like the model?	12	Similarity of each answer to gpt-4o-mini's answer to the same question	26 SE

Three definitions of "the same subject". Question 3 needs to know which texts are about the same thing. We use three definitions, from broad to exact:

- **Embedding topics.** All sampled texts of a corpus, across all quarters, are grouped into 50 clusters of similar vectors (k-means). A cluster is a topic; it means the same thing in every quarter. Available for all 30 corpora.
- **Stack Exchange tags.** Every Stack Exchange question carries one to five tags chosen by the asker from the community's vocabulary—for example `regression` and `bayesian` on Cross Validated, or `visas` and `schengen` on Travel. An answer inherits its question's tags, and two answers are on the same subject if their questions share at least one tag. Tags are defined by the community, not by us or by the embeddings.
- **The same question.** Two answers to the same question, the narrowest possible subject.

Subject measures. With subjects defined, each quarter has a *subject mix*: the share of its texts in each subject. We measure three things:

- *Same-subject similarity*: the average similarity over pairs of texts on the same subject. A rise means texts on the same subject become more alike.
- *Number of subjects*: the effective number of subjects, e^H, where H is the entropy of the subject mix. It reads as "how many equally common subjects this mix is worth": 50 if all 50 topics are equally common, 1 if every text is on one subject.
- *Subject-mix shift*: how different the quarter's subject mix is from the mix before ChatGPT (all pre-2022Q4 texts pooled), measured by the Jensen–Shannon divergence [5]: 0 for the same proportions, 1 for no subject in common. A rise means people now write about different things, in different proportions, than they used to.

The last two depend on how many texts a quarter has, and volumes fell after 2022 on several sites, so both are computed on random samples of 100 texts per quarter, averaged over 50 draws.

Model anchor. If people increasingly write like a language model, their answer to a question should increasingly resemble the model's answer to that same question. For up to 100 sampled answers per quarter we ask gpt-4o-mini to answer the same question (title and body) as a Stack Exchange user would, for every year—including years before ChatGPT existed, so the model acts as a fixed ruler. Some resemblance is expected simply because both texts are about the same subject, so we compare each human answer with the model's answer to its own question and with the model's answers to five *other* questions sharing a tag, from the same quarter. The difference, which we call *pull*, is the part of the resemblance specific to the question. In total 154,517 model answers were generated for 157,495 human answers.

4 Results

4.1 Answers became more similar to each other after ChatGPT

This section asks whether answers written in the same quarter resemble *each other* more than they used to (comparison with the model's own answers comes in Section 4.4). In 24 of 26 Stack Exchange communities the slope of answer-to-answer similarity increases after 2022Q4; the increase is significant in 15 (all 15 survive the multiple-testing correction) and in all 4 external corpora (Figure 2). The typical pattern is a flat or slowly declining similarity before ChatGPT that turns upward afterwards: 13 of the 19 significant corpora had a falling pre-2022 trend, 7 of them significantly. Two external corpora (PubMed, arXiv CS) were already rising before 2022; there the change is an acceleration rather than a reversal.

Is it just longer answers? Answers got longer after ChatGPT on several sites, and length alone can raise similarity: a long answer's vector is the average of many chunk vectors, and averaging smooths out what is specific to each chunk, pulling long answers towards a common, generic direction—so two long answers can look similar even if they say different things. To rule this out we repeat every test on exactly the first 120 tokens of each answer, so that all texts have the same length (shorter answers are left out). This gives the same verdict in 26 of 30 corpora; the four that differ (History, Astronomy, Bicycles, Photography) are significant *only* in the fixed-length version. Length does not drive the result.

Does it depend on the encoder? We repeated the test with a second, independently trained encoder, BGE-small [12]. In the 14 high-load communities and the 4 external corpora the two agree closely: 9 high-load communities and all 4 external corpora rise significantly with both. In the 12 low-load communities they do not: the second encoder finds a significant rise in only 2 and a significant fall in 3 (Seasoned Advice, Bicycles, Home Improvement). The rise is therefore robust in expert and academic writing, and encoder-dependent in everyday communities.



Figure 2: Slope change in answer-to-answer similarity after ChatGPT, per corpus. Circles: whole answers (filled = $p < 0.05$); diamonds: first 120 tokens only. Blue: high-load Stack Exchange; orange: low-load; green: external.

4.2 The turn is stronger where AI saves more effort

High-cognitive-load communities turn up more than low-load ones (Figure 3): the change is significant in 11 of 14 versus 4 of 12 (Fisher exact $p = 0.026$), and the median slope change, scaled by each site's pre-2022 variability, is 0.20 versus 0.09 (Mann–Whitney $p = 0.010$; permutation $p = 0.010$). High-load answers are somewhat longer, but answer length does not predict a site's slope change, and the difference holds with length as a control ($p = 0.026$). On the first 120 tokens alone more high-load sites are significant (13/14 vs 6/12, $p = 0.021$), but the effect sizes are closer ($p = 0.06$). With the second encoder the difference is larger: 9 of 14 versus 2 of 12 significant (Fisher $p = 0.019$), median scaled slope change 0.11 versus -0.02 (Mann–Whitney $p = 0.002$).

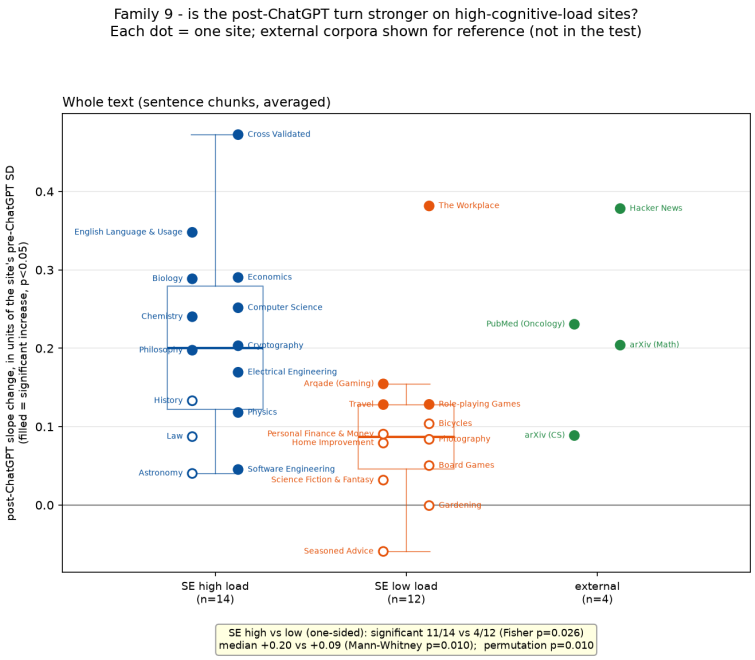


Figure 3: Slope change per Stack Exchange site, in units of its pre-ChatGPT variability; filled = significant. External corpora shown for reference, not part of the test.

4.3 What changed: the subjects and the questions, more than the answers

Answers can become more similar for three different reasons: people write about a narrower or different set of subjects; answers on the same subject converge; or answers to the same question converge. Table 3 tests each, using the three definitions of "the same subject" from the Method section.

Table 3: Number of corpora with a significant change after ChatGPT ($p < 0.05$).

Same subject defined as…	Out of	Same-subject texts converge	Subject mix shifts	Fewer subjects
Embedding topics	30	16	25	27
high / low / external	14 / 12 / 4	11 / 3 / 2	12 / 9 / 4	13 / 10 / 4
Stack Exchange tags	26	18	24	13
high / low load	14 / 12	9 / 9	12 / 12	8 / 5
The same question	26	6[a]	n/a[b]	n/a[b]
high / low load	14 / 12	4 / 2		

[a] 4 remain significant after the Benjamini–Hochberg correction. [b] Each question is asked once, so there is no mix of questions to compare across quarters.

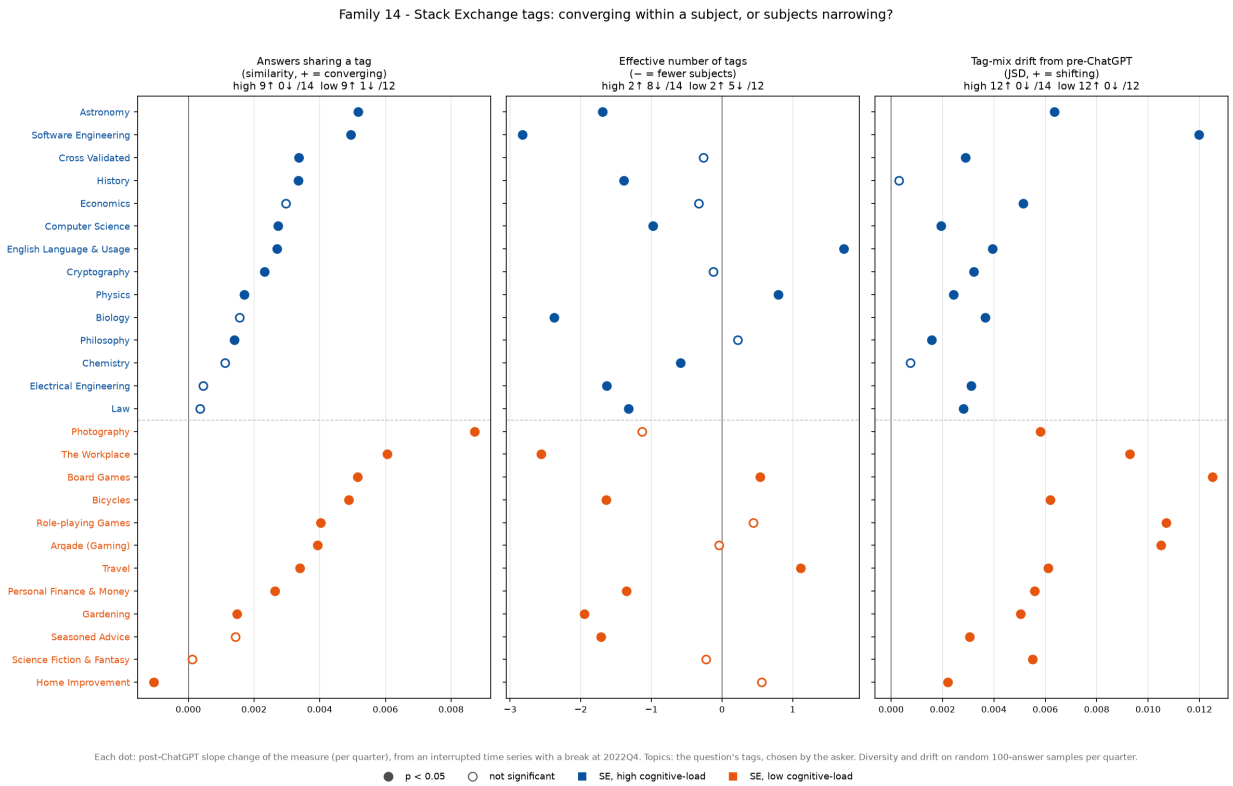


Figure 4: Subjects as Stack Exchange tags: per-site slope change in the similarity of answers sharing a tag (left), in the effective number of tags (centre) and in the shift of the tag mix from its pre-ChatGPT shape (right). Filled = $p < 0.05$.

Three findings follow:

- **What people discuss changes almost everywhere.** The subject mix moves away from its pre-ChatGPT shape in 24 of 26 sites by tags and 25 of 30 corpora by embedding topics, and the number of distinct subjects falls in most (Figure 4).
- **Answers on the same subject become more similar** in about two thirds of the sites by tags.
- **Answers to the exact same question mostly do not.** Only 6 of 26 sites show convergence, while any two answers from the same quarter, whatever their question, become more similar in 18 (Figure 5).

Read together: when the question is held fixed, people answer it no more alike than before. The convergence appears when the subject is broad, which means the *questions* within a subject have themselves become more similar—narrower, more standard, or concentrated on fewer issues. The change is in what is asked and discussed, more than in how people answer.

Two cautions. Embedding topics are found in the same vector space in which similarity is measured, so they can mistake answers that converge in wording for "fewer topics". Tags, chosen by askers, do not have this problem; that is why the two definitions disagree on low-load sites (3 versus 9 converging), and why we rely mainly on tags. And the number of distinct subjects falls on only half the sites by tags; on four it rises.

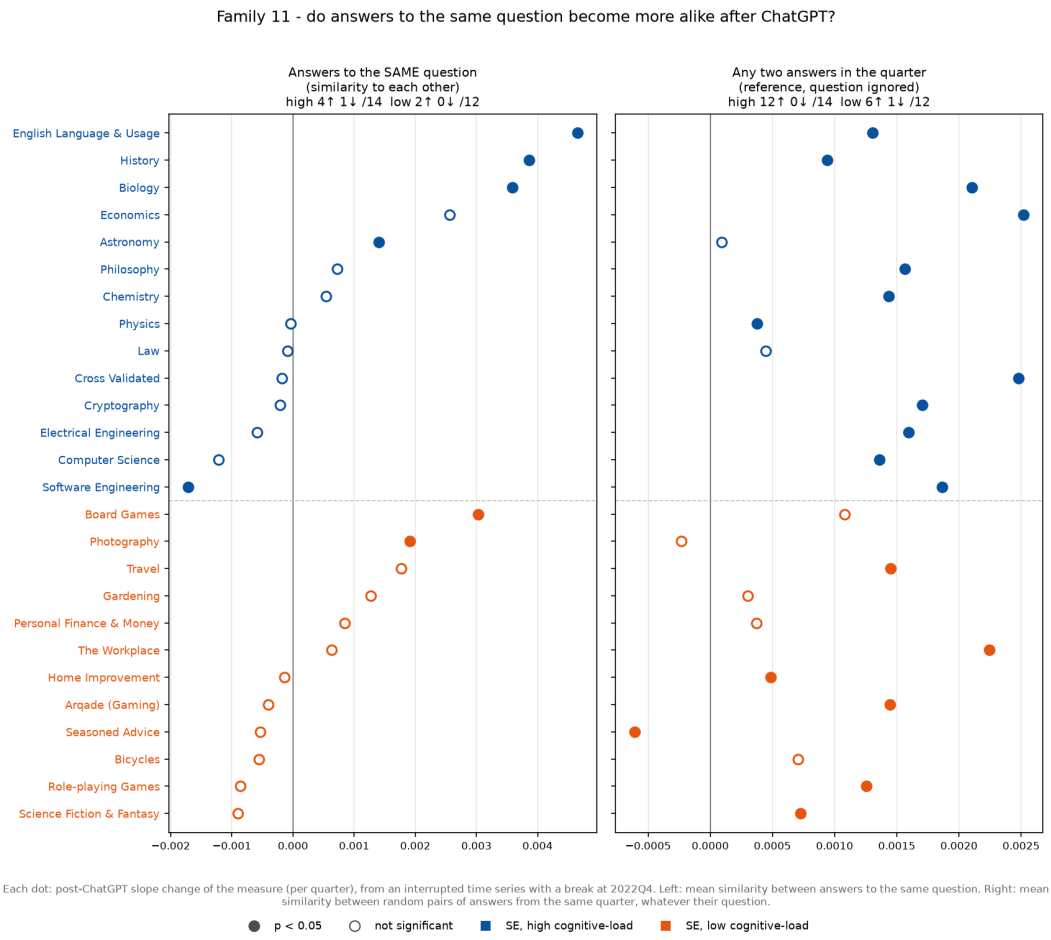


Figure 5: Left: similarity between answers to the same question. Right, for reference: similarity between random pairs of answers from the same quarter, whatever their question. Filled = $p < 0.05$.

4.4 People do not write more like the model

If people increasingly wrote like a language model, a human answer should increasingly resemble the model's answer to the same question—and more so than it resembles the model's answers to other questions on the same subject, which only reflect the shared subject. The test is sensitive: a human answer is much closer to the model's answer to its own question than to the model's answers to other same-tag questions (median similarity 0.64 versus 0.23). But that closeness does not grow after ChatGPT (Figure 6):

- similarity to the model's answer to the same question rises in 8 of 26 sites and falls in 2 (on the first 120 tokens: rises in 5, falls in 3);
- the question-specific part (*pull*) rises in 2 and falls in 6, none surviving the multiple-testing correction.

People's answers have not moved towards what the model says. The model's answers are stable: three independent answers to the same question have an average similarity of 0.92.

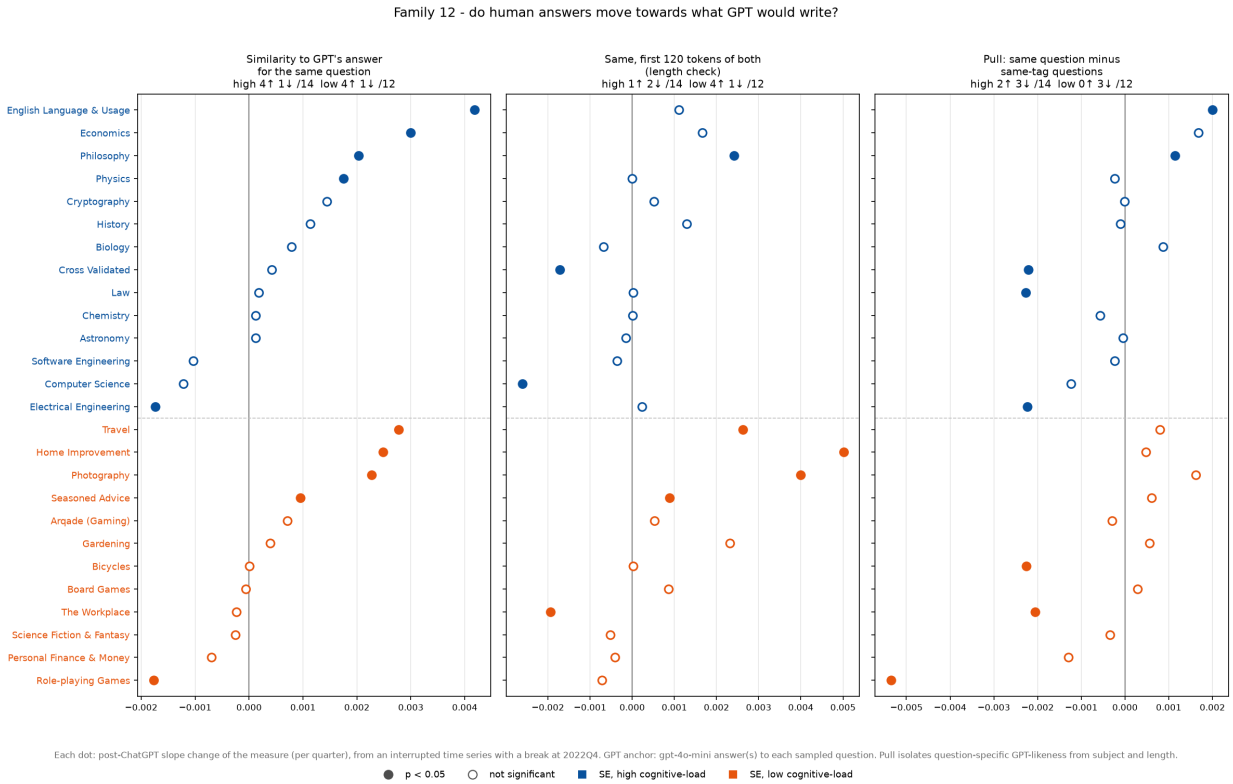


Figure 6: Per-site slope change in similarity to gpt-4o-mini's answer to the same question (whole text and first 120 tokens) and in pull (same question minus same-tag questions). Filled = $p < 0.05$.

4.5 Robustness

Table 4: Robustness checks.

Check	Result
Answer length	First 120 tokens only: same verdict in 26/30 corpora; the 4 differences are significant only at fixed length. Only each text's first K chunks (Cross Validated $K = 3$, PubMed $K = 4$): still significant.
Second encoder	BGE-small [12] on all 30 corpora: same verdict in 21/30 (whole text) and 20/30 (first 120 tokens); quarterly series correlate at a median of 0.66. High load: 11/14 agree (9 significant with both). External: 4/4. Low load: 6/12 agree; BGE finds 2 significant rises and 3 significant falls. High vs low (family 9): 9/14 vs 2/12, $p = 0.019$.
Number of topics	25 and 100 instead of 50 topics: the same conclusions (within-topic convergence 9–12/14 high vs 3–5/12 low; diversity falls in 24–26/30).
Sample size	Topic diversity and drift on fixed 100-text samples, as volumes fell after 2022: conclusions unchanged.
Model answers	One versus three answers per question (Economics): mean similarity between answers 0.92, so one is enough.

5 Discussion

The pattern is consistent across tests. Online answers became more similar to one another after ChatGPT—robustly in expert and academic communities, less clearly in everyday ones—mainly because the questions people bring and the subjects they discuss narrowed and shifted. People answering the same question do not converge, and they do not converge on the model's own answers.

One plausible mechanism is substitution: routine, well-defined questions that a chatbot can answer privately no longer reach public forums, which shifts and narrows what remains—especially in expert communities, where a model saves the most effort. That would explain why the subject mix moves almost everywhere, why same-subject answers converge, and why same-question answers do not. Our data are consistent with this reading but do not prove it.

This form of homogenization matters. A shrinking, shifting set of public questions is a narrowing of the public conversation itself: the long tail of unusual questions, and the record of how people reasoned about them, is what search engines, students and future models draw on. Because the change is measurable, it can be monitored. We maintain an open tracker that weekly collects Google's top results for 45 informational questions in five domains, fetches the full pages, and measures how much they repeat one another [3]; it began in October 2026 and will need several months before trends can be read.

Framing decision (open)

How strong a claim should this section make? From most to least cautious: (a) "a specific, measurable narrowing worth monitoring"; (b) "an early-warning sign of a narrowing public conversation"; (c) "evidence that AI is reshaping what people ask in public".

6 Limitations

- **Timing is not causation.** The break is fixed at ChatGPT's release; other changes happened in the same period (falling Stack Exchange volumes, platform policies, the pandemic, which is visible as a 2020 jump in Travel).
- **Small effects.** Slope changes are on the order of 10^{-3} per quarter; with many quarters, small effects become significant. They accumulate to a meaningful share of the between-quarter variation, but they are not large.
- **Encoder dependence in everyday communities.** The main results use one encoder. A second agrees in expert and academic corpora, but in low-load communities the rise depends on the encoder (Section 4.1); claims about everyday writing are therefore weaker.
- **Our own labels.** The high/low cognitive-load split is our classification of each community; the 26 sites are a small sample for a group comparison.
- **Low load is Stack Exchange only.** All external corpora are high-load.
- **The model anchor is a proxy.** gpt-4o-mini stands in for the different chatbots people actually used between 2022 and 2026.
- **Many tests.** We report counts after multiple-testing correction where it matters, and report results that do not support our reading (e.g. low-load embedding topics, model anchor).

7 Conclusion

Generative AI did not make everyone write the same way, and people do not write like the model. But online answers did become more similar to one another after ChatGPT—clearly in expert and academic communities, less clearly in everyday ones—because what people ask and discuss in public narrowed and shifted. That is a specific, measurable form of homogenization, and one worth watching.

A Exploratory analyses

Before the main analysis we ran several simpler measures. They are kept in the code and results (folder `artifacts/exploratory`) but superseded:

Family	Measure	Finding and why superseded
1	Lexical diversity and surface features	Rises after ChatGPT only because answers got longer; flat at fixed length [6].
2	Length-window sweep	Confirms family 1 at 50–200-token windows.
3	LSA (bag-of-words) similarity	Length artifact like family 1; replaced by sentence embeddings.
4	LDA topics	Topic mix nearly unchanged; replaced by embedding and tag topics.
6	BERTopic topics	Leaves about half of the texts unassigned, so within-topic results describe a selected half; replaced by families 13 and 14, kept for topic labels [4].
7	High- vs low-score answers	Both tiers rise (two sites); the rise is not confined to low-quality answers.

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